GIT QUEST | 30-DAY GIT LEARNING SERIES

Day 6: git diff, Spot the Difference

Week 1: First Steps | Level: Absolute Beginner

Created by Rutuja Naik

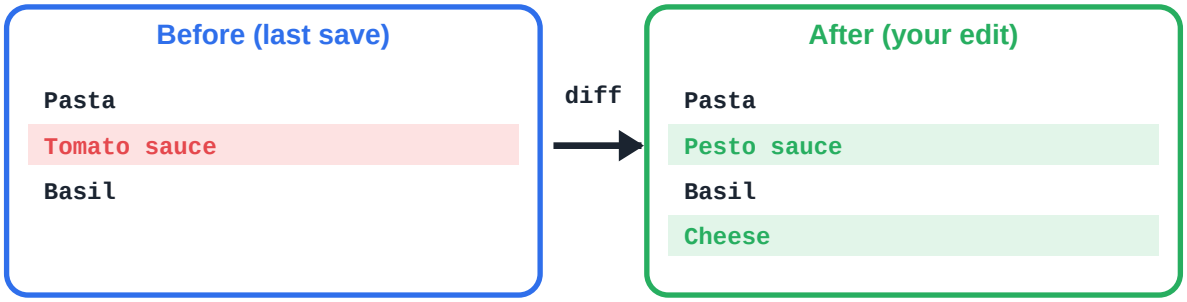
Today's mission

Yesterday you travelled back in time with `git log`. Today you learn to see **exactly what changed**. `git diff` is the spot-the-difference puzzle of Git.

- Use `git diff` to see changes you have not staged yet
- Use `git diff --staged` to see what is about to be committed
- Compare two commits
- Read a diff: which lines were removed and which were added

1. The big idea: red pen and green pen

Imagine an editor checking your homework with two pens. The **red pen** crosses out what was **removed**. The **green pen** writes in what was **added**. `git diff` does exactly that for your files.



Command	Compares	Question it answers
git diff	Desk vs packing table	What did I change but not pack yet?
git diff --staged	Packing table vs cupboard	What am I about to commit?
git diff HEAD	Desk vs cupboard	Everything since my last save
git diff A B	Two save points	What changed between these commits?

2. Reading a diff

```
diff --git a/recipe.txt b/recipe.txt
--- a/recipe.txt
+++ b/recipe.txt
@@ -1,3 +1,4 @@
 Pasta
-Tomato sauce
+Pesto sauce
 Basil
+Cheese
```

Part	Meaning
--- a/recipe.txt	The old version (what Git last saved)
+++ b/recipe.txt	The new version (what is on your desk)
@@ **-1,3 +1,4** @@	Location: the old file showed 3 lines from line 1, the new one shows 4
-Tomato sauce	RED: this line was removed
+Pesto sauce	GREEN: this line was added
(space) Pasta	Unchanged line, shown for context

> **Good to know**
> A changed line always appears as one **-** line and one **+** line: the old text out, the new text in.

3. Step by step

Step 1: Save a tiny recipe

```
$ git init -b main spot-diff
$ cd spot-diff
$ printf "Pasta\nTomato sauce\nBasil\n" > recipe.txt
$ git add recipe.txt
$ git commit -m "Add recipe"
```

Step 2: Change it and spot the difference

```
$ printf "Pasta\nPesto sauce\nBasil\nCheese\n" > recipe.txt
$ git diff
-Tomato sauce
+Pesto sauce
+Cheese
```

Step 3: Pack it, then look again

```
$ git add recipe.txt
$ git diff
(nothing: the changes moved to the packing table)
```

Step 4: Check what you are about to commit

```
$ git diff --staged
-Tomato sauce
+Pesto sauce
+Cheese
```

Step 5: Commit and compare two saves

```
$ git commit -m "Switch to pesto and add cheese"
$ git diff --stat HEAD~1 HEAD
 recipe.txt | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)
```

Bonus: word-level diff

```
$ git diff --word-diff
[-Tomato-]{+Pesto+} sauce
{+Cheese+}
```

Common problems

You see	What it means and the fix
git diff prints nothing	Changes are already staged, or nothing changed. Try git diff --staged
A new file is missing	Untracked files are not shown. Run git add, then git diff --staged
Screen stuck with (END)	Press q to quit the scrolling screen

4. Cheat sheet

Command	What it does
git diff	Desk vs packing table
git diff --staged	Packing table vs cupboard (what you will commit)
git diff HEAD	Desk vs cupboard (all uncommitted changes)
git diff HEAD~1 HEAD	Compare the last two commits
git diff --stat	Short summary of changed files
git diff <file>	Show changes in one file only
git diff --word-diff	Highlight changed words

5. Today's 3 commits

#	File	Content	Commit message
1	day06-git-diff/lesson.md	Today's lesson: red pen, green pen	docs(day06): add lesson on git diff
2	day06-git-diff/demo.tape	VHS script that records the terminal GIF	feat(day06): add VHS tape for git diff demo
3	exercise.md and days-1-to-6-summary.md	Challenge, plus a recap of Days 1 to 6	docs(day06): add exercise and days 1-6 summary

6. Mermaid diagram GitHub draws for you

```
```mermaid
flowchart LR
  A["Working directory"] -- "git diff" --> B["Staging area"]
  B -- "git diff --staged" --> C["Repository (HEAD)"]
  A -. "git diff HEAD" .-> C
```
```

7. Challenge and quiz

Challenge: change one line and add one line in a saved file. Predict what `git diff` and `git diff --staged` will show, then check yourself.

Q1. Which command shows changes you have already staged?

Q2. What does a line starting with - mean?

Q3. Why is git diff empty right after git add?

Answers: 1) git diff --staged. 2) The line was removed. 3) The changes moved to the packing table.

> **Coming up: Day 7**
> **Week 1 recap, quiz and cheat sheet.** Lock in everything from Days 1 to 6.